Adjustment and external replication of drug–hearing-loss signals: a disproportionality analysis of FAERS and Canada Vigilance

Oleg Tropynin
Independent researcher, Karaganda, Kazakhstan
tropininoleg@gmail.com

Abstract

Introduction: Recent disproportionality analyses of the US Food and Drug Administration Adverse Event Reporting System (FAERS) reported many drug–hearing-loss signals absent from labels. We assessed which survive adjustment for hearing-specific biases and external replication.
Methods: Disproportionality analysis of 19.9 million deduplicated FAERS reports (2004–2026) and 1.2 million Canada Vigilance reports, following a protocol committed before analysis. For hearing loss and tinnitus, crude reporting odds ratios (signal: lower 95% bound > 1, ≥ 3 cases) were compared with Mantel–Haenszel estimates stratified by age, sex and period after excluding ear-related indications, reports naming established ototoxic drugs and lawyer reports. Methods were compared on a label-derived reference set (area under the receiver operating characteristic curve, AUC) and by replication in Canada Vigilance; robust unlabelled signals underwent case-by-case review.
Results: Adjustment changed the hearing-loss signal list (300 crude, 310 adjusted, 63 lost) but not agreement with labels (AUC 0.718 vs 0.723). Unlabelled signals replicated in Canada more often than background but less often than labelled ones; power-calibrated (post hoc), labelled signals replicated as predicted (observed/expected 1.02, 95% CI 0.77–1.32) and unlabelled ones did not (0.44, 0.29–0.62), also within strata of consumer reporting. Solicited reports, report clusters and shared co-medication produced spurious replications. Of 22 robust unlabelled signals (post hoc), 15 had artefactual or known explanations; 7 remained (2–9 under alternative choices), led by pregabalin.
Conclusions: Most unlabelled FAERS hearing-loss signals do not survive adjusted replication and provenance review. These findings are hypothesis-generating; cross-database replication needs adjusted estimates and explicit handling of solicited reports.

Key points

- Adjusting FAERS hearing-loss screens for indication, ototoxic co-medication and stimulated reporting changes which drugs are flagged but not their agreement with drug labels.
- In Canada Vigilance, labelled hearing-loss signals replicated about as often as predicted and unlabelled ones less than half as often, also among drugs with similar consumer reporting; solicited reports and shared co-medication produced spurious replications.

- After adjusted replication and case-by-case review of report provenance, 7 of 22 robust unlabelled signals remained, led by pregabalin; they are hypotheses for exposure-based studies.

1 Introduction

Drug-induced hearing loss ranges from reversible (salicylates, loop diuretics, quinine, macrolides) to permanent (aminoglycosides, platinum agents), and the classical ototoxic drug classes were identified mainly through clinical observation and case series [1]. For newer drugs, post-marketing spontaneous reports are often the first source of evidence, and recent studies have screened the FDA Adverse Event Reporting System (FAERS) across all drugs for hearing-loss signals [2–4] and for tinnitus [5, 6]; national databases in Japan and France have been used for the same purpose [7, 8].

The FAERS screens share a design: crude disproportionality across all reports, with novelty defined as absence from the label. In this setting crude disproportionality is open to biases specific to hearing outcomes. Corticosteroids, vasodilators and antivirals are prescribed *for* sudden hearing loss and tinnitus, and several diseases cause hearing loss themselves, so their treatments co-occur with the event (confounding by indication). A drug routinely given with a strongly ototoxic one—an antiemetic with cisplatin, an antibiotic with an aminoglycoside—inherits part of the association (co-medication or "innocent bystander" bias). Reporting can be stimulated by safety communications, litigation or media attention [9]. Editorials have questioned the value of disproportionality analyses that do not address such biases [10], and few FAERS screens test their signal lists against an independent data source or a reference set, as done in evaluations of signal-detection methods [11].

We therefore asked a narrower question than "which drugs are ototoxic?": *which hearing-loss and tinnitus signals remain after adjustment for these biases, and do the signals that remain replicate in an independent national database and over time?* We compared crude and adjusted detection against a reference set derived from US labels, replicated every signal in Canada Vigilance, re-tested the signals reported by previous FAERS screens, and reviewed the report provenance of every robust unlabelled signal.

2 Methods

The study is reported according to the READUS-PV guideline [12, 13]; the checklist is given in Supplementary Table S1. The analysis protocol, with all definitions and thresholds of the primary analysis, was written and committed to a version-controlled repository (git commit ff77c40, 26 October 2026, 02:05 UTC+5) before the first result was computed; it is time-stamped, not registered with a third party; every later change is dated and labelled in that file (`PROTOCOL.md`), and analyses added after results were seen are reported as post hoc.

2.1 Data sources

We used all 90 quarterly ASCII extracts of the legacy Adverse Event Reporting System (2004Q1–2012Q3) and FAERS (2012Q4–2026Q2), published by the US Food and Drug Administration; the Canada Vigilance Adverse Reaction Online Database extract (Health Canada; reports received 1965 to 30 June 2026); and the openFDA bulk export of structured product labels (23 October 2026) [14]. Events are coded in MedDRA preferred terms (PTs), at the version current when each quarter was released (FAERS) and version 29.0 (Canada Vigilance); outcome terms were matched by preferred-term name, which is stable across these versions for the terms used. All data are public and de-identified.

2.2 Pre-processing and deduplication

In FAERS we kept the most recent version of each case (by FDA receipt date, preferring FAERS over legacy records, then case version), removed cases on FDA deletion lists, removed report identifiers recorded under two case numbers, and collapsed content duplicates: reports with identical sex, age, event date, country, set of suspect ingredients and set of reactions (applied only when event date and age were present). Drug names were mapped to active ingredients: the `prod_ai` field was split into ingredients; salt and hydrate words were removed when the remaining base was itself an observed ingredient; free-text names in records without `prod_ai` were mapped to the dominant ingredient recorded for that name ($\geq$70% of $\geq$5 records); INN/USAN synonyms were merged (for example glycopyrronium and glycopyrrolate); four-letter biologic suffixes were removed; and non-specific entries (for example "vitamins", "minerals", "... NOS") were dropped. Combination products contribute to each of their ingredients. In Canada Vigilance, reports flagged as duplicates of a report present in the extract were removed and product ingredients were normalised with the same rules.

2.3 Outcomes, exposure and comparator

The primary outcome, hearing loss, comprised the MedDRA PTs Deafness; Deafness unilateral, bilateral, neurosensory, transitory, permanent and mixed; Hypoacusis; Neurosensory hypoacusis; Sudden hearing loss; Ototoxicity; Hearing impaired; Hearing disability; and High frequency hearing loss. Conductive and congenital deafness were excluded. The secondary outcome was Tinnitus. An ingredient was exposed when reported as a suspect drug (FAERS primary suspect, secondary suspect or interacting; Canada suspect or interacting). The comparator for each ingredient was all other reports in the same population. Missing age and sex formed their own strata; missing indication or country was treated as not meeting the corresponding exclusion.

2.4 Disproportionality

For each ingredient we computed the reporting odds ratio (ROR) with a Woolf 95% confidence interval [15, 16] and the information component (IC) with its lower credibility bound IC_{025} [17]. A

crude signal required at least three cases, a lower ROR bound above 1 and $IC_{025} > 0$.

The pre-specified *adjusted* analysis was run in a restricted population and stratified. We excluded (i) reports in which any drug had an ear or hearing indication (deafness, hearing, tinnitus, ototoxicity, Ménière's disease, otitis, vertigo, vestibular, labyrinthine and cochlear terms); (ii) reports that mentioned, in any role, a drug from an a-priori list of 29 drugs from the established ototoxic classes [1] (aminoglycosides, capreomycin and vancomycin; cisplatin, carboplatin and oxaliplatin; loop diuretics; quinine and related antimalarials; aspirin; macrolides; deferoxamine; vincristine; eflornithine) other than the drug under study; and (iii) FAERS reports submitted by lawyers (Canada Vigilance does not record this). For a drug on the list, reports naming the drug itself in a non-suspect role were also removed from its population. We then computed the Mantel–Haenszel (MH) OR [18] over 60 strata (five age bands × sex [male, female, unknown] × four periods of report receipt: before 2004, 2004–2012, 2013–2018, 2019–2026) with the Robins–Breslow–Greenland variance [19]. An *adjusted signal* required at least three cases in the restricted population and a lower 95% bound above 1. The implementation reproduces `statsmodels` estimates on simulated data to nine decimal places.

2.5 Reference set

From single-ingredient US labels we classified each ingredient as label-positive when a hearing-loss term appeared in a safety section (boxed warning, warnings, precautions, adverse reactions, post-marketing experience, contraindications, overdosage, drug abuse and dependence, or the corresponding over-the-counter sections) of at least half of its prescription labels, and label-negative when no label mentioned hearing loss or tinnitus; other ingredients were left out. Phrases that contain a hearing word without describing an adverse effect (for example the disease name "maternally inherited diabetes and deafness" and telephone lines for deaf users) were removed first. Discrimination was measured by the area under the ROC curve (AUC) among ingredients with ≥1,000 suspect FAERS reports, comparing the crude and adjusted lower 95% bounds (the like-for-like pair; the protocol pre-specified the crude IC_{025}, which is also reported) and point estimates; the difference was bootstrapped (2,000 resamples). Because labels are partly informed by spontaneous reports, the reference set compares methods with each other; it does not estimate absolute accuracy.

2.6 Replication

For external replication, signals were derived in FAERS after removing reports of Canadian origin and tested in Canada Vigilance. Pre-specified rule: an ingredient was testable when the observed or expected number of Canadian cases was at least three, and replicated when the Canadian lower ROR bound exceeded 1 with at least three cases. The same rule applied to label-negative ingredients that were neither crude nor adjusted FAERS signals gave the background rate of replication without a known association; label-negative ingredients that *were* adjusted signals form a second reference group. Because the testability rule depends on the observed count, we also computed a power-calibrated comparison (post hoc): for every ingredient with a Canadian expected count $E > 0$,

replication was defined as an observed count of at least a_{crit}, the smallest count ≥ 3 with Poisson $P(X \geq a \mid E) < 0.025$, and the expected number of replications was $\sum_i P(X \geq a_{\text{crit},i} \mid E_i \times \text{OR}_i)$, with OR_i the adjusted FAERS estimate shrunk towards the null by empirical Bayes (log OR $\times \tau^2/(\tau^2 + \text{se}^2)$, with τ^2 estimated across all ingredients) to counter the selection of signals on their estimate; observed-to-expected ratios near 1 indicate that signals replicate as often as their FAERS strength predicts. Confidence intervals were bootstrapped over ingredients (2,000 resamples). In the post-hoc adjusted-to-adjusted analysis the Canadian adjusted case count was compared with its stratified expected count. For temporal replication, signals derived from reports received up to 2014 were tested in reports received from 2015 onwards. Rates were compared with Fisher's exact test.

2.7 Post-hoc and sensitivity analyses

After the primary results were seen we added: Canada Vigilance without solicited reports ("Study" report type, which includes manufacturers' patient-support programmes); an extended indication filter (further ear terms; diseases that themselves cause hearing loss, such as Fabry disease, mucopolysaccharidoses, cryopyrin-associated syndromes, meningitis, giant cell arteritis and thyroid eye disease; and reports of prenatal exposure); and replication in which the Canadian estimate is adjusted in the same way as the FAERS one (*adjusted-to-adjusted*). In this post-hoc framework, ototoxic co-medication was handled in three further ways: as a stratum dimension instead of an exclusion (because restricting on a variable that reporters may add after observing hearing loss can introduce selection bias), with a short list of strongly ototoxic drugs only, and after excluding reports that list 15 or more drugs (report clusters). Canada Vigilance was also restricted to reports received from 2004.

2.8 Robust signals and case-level review

A *robust signal* (post hoc) was an adjusted FAERS signal that replicated adjusted-to-adjusted in Canada Vigilance without solicited reports. Throughout, a signal is *labelled* when the event is mentioned in any safety section of any US label of a product containing the drug (single-ingredient or combination) and *unlabelled* otherwise, including drugs with no US label in the export; *label-positive* and *label-negative* refer to the stricter reference-set classes. Robust unlabelled signals were reviewed. For each we profiled the FAERS case series (without Canadian reports): share of cases from the most frequent country and receipt year, consumer reports, reports listing $\geq$15 drugs, reports co-naming an established ototoxic drug, prenatal exposure and leading indications; we searched PubMed for the drug with hearing terms. Each signal was assigned the first applicable category of a written rubric: K, established ototoxic class or association documented by a prospective or cohort clinical series; A, report cluster ($\geq$50% of cases in reports listing $\geq$15 drugs); B, ototoxic co-medication ($\geq$50% of cases co-name an established ototoxic drug); C, stimulated or solicited reporting documented for the drug; D, main indication itself associated with the event; E, none of these. The rubric distinguishes C from D only on documented evidence; a high consumer share alone does not make

a signal C. For category D we additionally required that the signal not persist against other drugs used for the same indication: within FAERS reports (without Canadian origin) whose indication matched the drug's main indication, we computed the ROR of the drug against all other suspect drugs (active-comparator check, post hoc). The author applied the rubric, and a second, AI-based assessor applied it blind to the author's assignments, with the same profiles and literature access; agreement is reported as Cohen's κ.

2.9 Software and use of AI tools

Analyses used Python 3.10 (numpy 2.2.6, pandas 2.3.3, scipy 1.15.3, statsmodels 0.15.0, matplotlib 3.10.9). Data processing code, analyses and the first draft of this manuscript were produced with an AI assistant (Claude, Anthropic) working under the author's direction; three AI reviewers (methods, clinical, editorial) critiqued the draft, and their findings were addressed as described in the protocol file. The author reviewed the code, results and text and is responsible for them.

3 Results

3.1 Data

The 90 FAERS extracts held 24,812,424 report versions, from which 20,692,682 cases remained after keeping the latest version; 104,186 cases on FDA deletion lists, cases without a reaction or mappable ingredient, 1,681 report identifiers recorded under two cases and 510,312 content duplicates were then removed, leaving **19,876,330 reports** (4% of drug rows could not be mapped; Supplementary Figure S1). They contained 83,883 hearing-loss and 42,900 tinnitus reports; hearing-loss reports came more often from consumers and from patients aged $\geq$65 years than other reports (Supplementary Table S4). Without Canadian reports, the population used for signal detection held 19,135,329 reports and 8,122 suspect ingredients; its adjusted subset excluded 18,735 reports with an ear or hearing indication, 1,712,914 reports naming a listed ototoxic drug and 484,957 lawyer reports, leaving 16,962,022 reports (67,913 hearing-loss cases). Canada Vigilance contributed 1,210,857 reports after removing 58,623 duplicates (4,000 hearing-loss and 3,284 tinnitus reports).

Drugs of the established ototoxic classes gave strong signals in both databases (Supplementary Table S2), for example amikacin (FAERS adjusted OR 12.7 (11.6–13.9)), gentamicin (11.4 (9.8–13.2)) and cisplatin (4.3 (4.0–4.6)); so did the PDE5 inhibitors sildenafil (3.9 (3.7–4.2)) and tadalafil (4.2 (3.9–4.6)), whose labels carry hearing-loss warnings, and teprotumumab (59.8 (55.4–64.5)). Some label-positive drugs did not signal: oxaliplatin (crude ROR 0.44) and vardenafil (6 cases); carboplatin was a crude signal that fell below 1 after adjustment (1.96 to 0.82), as did vincristine's estimate, an example of the sensitivity that the co-medication filter costs.

3.2 Adjustment changes the signal list but not its agreement with labels

In FAERS without Canadian reports the crude rule flagged 300 ingredients for hearing loss and the adjusted rule 310; 63 crude signals were lost and 73 appeared (Table 1). Against the reference set (106 label-positive and 927 label-negative ingredients with ≥1,000 suspect reports), discrimination was moderate and practically identical: crude IC_{025} versus adjusted lower bound, as pre-specified, AUC 0.720 vs 0.723; crude versus adjusted lower bound, the like-for-like pair, 0.718 vs 0.723 (difference 0.005, 95% CI -0.023 to 0.030). The same held for point estimates (0.700 vs 0.705), after excluding the 83 ingredients without an estimable adjusted bound (0.698 vs 0.703), at lower report thresholds, without the 29 listed drugs, and for tinnitus (0.775 vs 0.777). Adjustment lowered sensitivity (0.40 to 0.35) with little change in specificity (0.866 to 0.873).

3.3 Replication in Canada Vigilance

Under the pre-specified rule, adjusted hearing-loss signals already on a label replicated in 34/50 (68%) of testable cases, unlabelled adjusted signals in 31/76 (41%), label-negative adjusted signals in 21/54 (39%), crude-only signals in 9/21 (43%), and label-negative ingredients that were neither crude nor adjusted signals in 43/214 (20%) (all label-negative ingredients, as literally specified in the protocol: 68/281 (24%); Table 1). In this pre-specified comparison unlabelled signals replicated more often than the background (odds ratio 2.74, Fisher $p < 0.001$) but less often than labelled signals. The power-calibrated comparison (post hoc) placed this contrast on a common scale (Figure 1): labelled signals (n=70 ingredients) replicated about as often as their shrunken FAERS estimates predicted (31 observed vs 30.5 expected; O/E 1.02, 95% CI 0.77–1.32), whereas unlabelled signals (n=201) replicated less than half as often (25 observed vs 56.8 expected; O/E 0.44, 95% CI 0.29–0.62); the ratio of the two was 0.43 (95% CI 0.26–0.66). The label-negative signals, a subset of the unlabelled group, behaved alike (18 observed vs 37.5 expected; O/E 0.48, 95% CI 0.27–0.72). Results were the same with unshrunken estimates, with the pre-specified label definition, and for tinnitus (60 observed vs 63.2 expected; O/E 0.95, 95% CI 0.79–1.14 for labelled and 22 observed vs 46.4 expected; O/E 0.47, 95% CI 0.29–0.66 for unlabelled signals).

Unlabelled signals were more often reported by consumers (median share of FAERS cases 58% vs 39% for labelled signals) and more often from the United States (67% vs 50%), so provenance could explain the contrast. It did not: among signals with a consumer share at or below the median, O/E was 1.01 for labelled and 0.57 (95% CI 0.27–0.93) for unlabelled signals, and above the median 1.04 and 0.36. Pooled across ingredients (random effects), the Canadian ROR was 3.60 (2.47–5.26) for labelled and 1.59 (1.28–1.96) for unlabelled signals: as a group, unlabelled signals were attenuated rather than absent.

The replication test was not conservative: label-negative non-signals replicated 4.2 times as often as expected without any association (36 vs 8.5). Solicited reports explained much of this: *Study* reports, which include manufacturers' patient-support programmes, made up 61% of Canadian hearing-loss cases for the background drugs that replicated, against 14% for replicated labelled signals and 36% overall; for ocrelizumab, mepolizumab and omalizumab the share was 93%, 93%

and 86%, against 1% for gentamicin and 1% for cisplatin. The mirror image appeared in FAERS: tafamidis (adjusted OR 4.3 (3.8–4.8)) had 361 hearing-loss reports in FAERS (86% from consumers, 97% from the United States) but none in Canada Vigilance against 3.4 expected, and sacubitril (2,801 FAERS hearing-loss reports; 88% from consumers) none against 1.9 expected (too few to be testable).

Co-medication bias can also replicate when crude estimates are compared. Etoposide, given in cisplatin-containing regimens, was a crude signal in both databases (FAERS ROR 2.09 (1.91–2.28); Canada 5.13 (3.68–7.14)); after adjustment the FAERS estimate reversed (0.67 (0.46–0.98)) and the Canadian one rested on 2 cases (1.15 (0.29–4.64)). Temporal replication within FAERS did not favour adjustment for hearing loss: early crude-only signals recurred after 2014 at least as often (18/28 (64%)) as early adjusted signals (81/155 (52%); background 137/938 (15%)), consistent with biases that persist within one database; for tinnitus adjusted signals recurred more often (100/121 (83%) vs 9/15 (60%)).

3.4 Post-hoc analyses

With the extended indication filter, Canada Vigilance restricted to unsolicited reports and both estimates adjusted (Table 2), labelled hearing-loss signals replicated in 26/44 (59%), unlabelled signals in 10/51 (20%) and the background in 19/158 (12%) (Fisher $p = 0.24$); calibrated on adjusted Canadian counts, O/E was 1.10 (95% CI 0.79–1.48) for labelled and 0.26 (0.11–0.43) for unlabelled signals (ratio 0.24, 95% CI 0.09–0.44); the pooled Canadian adjusted OR was 4.25 (2.73–6.64) for labelled, 1.37 (1.03–1.83) for unlabelled and 1.29 (1.00–1.66) for label-negative signals. For tinnitus, labelled signals replicated less fully in this framework (O/E 0.70, 95% CI 0.54–0.88), with unlabelled signals lower still (0.21). Across the co-medication variants and the restriction of Canada Vigilance to reports from 2004, unlabelled O/E ranged 0.23–0.28 and labelled 0.94–1.18. The comparison with the background depended on report clusters: after excluding reports that list $\geq$15 drugs, the background fell to 5/159 (3%) and unlabelled signals (8/48 (17%)) exceeded it (Fisher $p = 0.002$), although still replicating far below expectation (O/E 0.27). Handling co-medication as a stratum rather than an exclusion removed the below-1 estimates of label-positive drugs (carboplatin 0.79 to 1.34; vincristine 0.48 to 1.04), consistent with selection bias from the exclusion, but did not remove the etoposide association (1.57), consistent with residual co-medication bias under a binary stratum.

3.5 Signals from previous screens

Of the 23 hearing-loss signals that Li et al. classified as unlabelled [2], 16 were crude and 16 adjusted signals in FAERS without Canadian reports, 6 were testable in Canada and 2 replicated; 1 met the post-hoc robust definition (glycopyrrolate). Their strongest unlabelled signals included sacubitril/valsartan and tafamidis. Of the 28 unlabelled paediatric signals of Liu et al. [4], 11 were adjusted signals in our all-age data and 1 met the robust definition (azithromycin, which is on

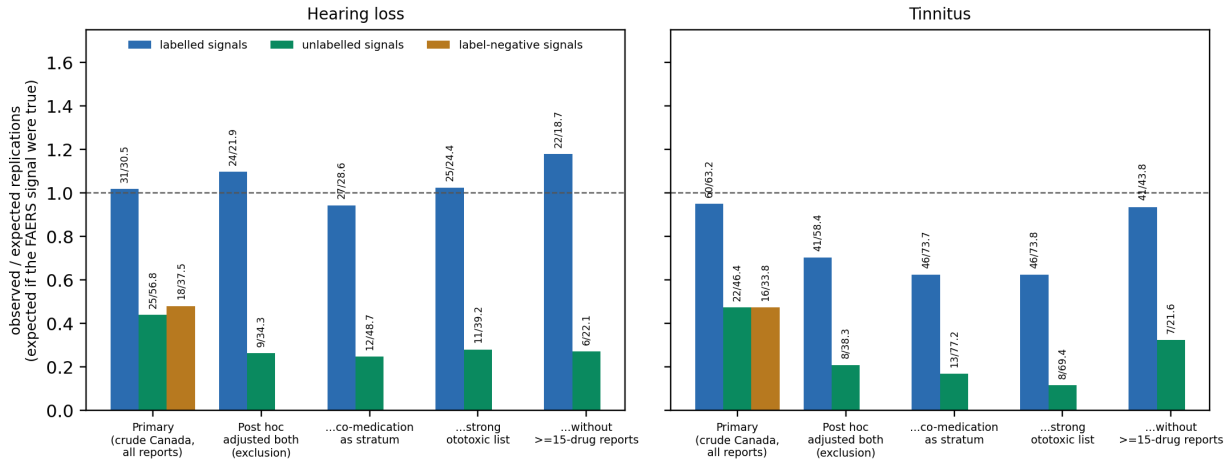


Figure 1: Power-calibrated replication in Canada Vigilance: observed replications divided by those expected if each FAERS adjusted estimate were true (labels give observed/expected counts). Primary: pre-specified signals, Canada crude estimates, all reports. Post hoc: both databases adjusted, Canada without solicited reports, four ways of handling co-medication. A ratio near 1 means signals replicate as their FAERS strength predicts.

the a-priori ototoxic list); cefixime, ceftriaxone, etoposide and isoniazid were lost on adjustment (Table 3). Only abstract-level lists were available for the other screens (Supplementary Table S6).

3.6 Case-level review

Under the post-hoc definition, 10 hearing-loss and 12 tinnitus signals were robust and absent from every US label (Table 4, Figure 2). The review attributed 4 to report clusters (for example candesartan–tinnitus: 80% of cases from the United Kingdom, median 42 drugs per report), 6 to stimulated or mass-exposure reporting (mRNA COVID-19 vaccines, rofecoxib, finasteride), 1 to an established ototoxic class (oral paromomycin) and 4 to an indication, with an active-comparator estimate consistent with it (etanercept against other rheumatoid-arthritis drugs, ROR 0.93 (0.87–1.00); insulin aspart within diabetes, 1.17 (1.00–1.37); galcanezumab within migraine, 0.82 (0.63–1.06); and chlorthalidone within hypertension, 1.04 (0.43–2.51), which is indeterminate with 5 cases). For seven signals the indication did not account for the association, because it persisted against other drugs for the same indication: tofacitinib, pregabalin, amlodipine, glycopyrrolate and travoprost for hearing loss (for example pregabalin within pain and neuropathy, ROR 3.27 (3.04–3.52); amlodipine within hypertension, 1.89 (1.69–2.10)) and doxycycline and liothyronine for tinnitus (doxycycline, 3.63 (2.98–4.42)). A second, AI-based assessor applying the same rubric to the same profiles agreed on 19 of 22 assignments before the active-comparator step (Cohen's $\kappa = 0.82$) and on 19 after the same mechanical rule was applied to both ($\kappa = 0.82$); with its assignments, nine signals would remain unexplained (Supplementary Table S3). Before the post-hoc active-comparator rule, only two (pregabalin and doxycycline) were unexplained; the residual count therefore ranges from 2 to 9 depending on post-hoc choices and assessor.

Table 1: Crude versus adjusted detection (pre-specified analyses). Replication counts are replicated/testable ingredients.

	Crude	Adjusted
Hearing loss		
Reference set, positive/negative ingredients	106/927	
AUC, lower 95% bound	0.718	0.723
difference (95% CI)	0.005 (-0.023 to 0.030)	
Sensitivity / specificity of the signal rule	0.40 / 0.87	0.35 / 0.87
Signals (FAERS without Canadian reports)	300	310
Replicated in Canada[a]: labelled adjusted signals	34/50 (68%)	
unlabelled adjusted signals	31/76 (41%)	
label-negative adjusted signals	21/54 (39%)	
crude-only signals	9/21 (43%)	
background (label-negative non-signals)	43/214 (20%)	
all label-negative ingredients (protocol wording)	68/281 (24%)	
Observed/expected replications[b]: labelled	31/30.5 = 1.02 (0.77–1.32)	
unlabelled	25/56.8 = 0.44 (0.29–0.62)	
label-negative	18/37.5 = 0.48 (0.27–0.72)	
Tinnitus		
Reference set, positive/negative ingredients	186/927	
AUC, lower 95% bound	0.775	0.777
difference (95% CI)	0.002 (-0.016 to 0.018)	
Sensitivity / specificity of the signal rule	0.51 / 0.85	0.50 / 0.87
Signals (FAERS without Canadian reports)	399	349
Replicated in Canada[a]: labelled adjusted signals	65/89 (73%)	
unlabelled adjusted signals	24/56 (43%)	
label-negative adjusted signals	17/43 (40%)	
crude-only signals	13/30 (43%)	
background (label-negative non-signals)	31/193 (16%)	
all label-negative ingredients (protocol wording)	53/252 (21%)	
Observed/expected replications[b]: labelled	60/63.2 = 0.95 (0.79–1.14)	
unlabelled	22/46.4 = 0.47 (0.29–0.66)	
label-negative	16/33.8 = 0.47 (0.27–0.72)	

[a] Pre-specified rule: FAERS without Canadian reports; Canada crude ROR, all reports. [b] Power-calibrated (post hoc): observed replications vs those expected if each FAERS adjusted estimate were true.

4 Discussion

Three findings matter for anyone screening spontaneous reports for ototoxicity. First, adjusting for indication, ototoxic co-medication, lawyer reports and demographics changed a substantial part of the hearing-loss signal list (63 of 300 crude signals lost, 73 gained) but did not improve agreement with labels. Second, replication in an independent national database behaved as a validation test should for labelled hearing-loss associations—they replicated about as often as their strength predicted—whereas unlabelled signals replicated less than half as often, a contrast that held within strata of consumer reporting and whose pooled Canadian estimate (attenuated, not absent) was close to that of signals for drugs whose labels are silent on hearing. Third, replication is not a clean validator on its own: solicited patient-support-programme reports and report clusters pro-

Table 2: Post-hoc analyses: extended indication filter, Canada Vigilance without solicited reports, both estimates adjusted, with four ways of handling ototoxic co-medication and a restriction of Canada Vigilance to reports from 2004. Fisher p: unlabelled vs background replication. O/E: observed/expected replications (adjusted counts). Robust: adjusted signals replicated adjusted-to-adjusted (unlabelled in brackets).

Co-medication handling	Signals	AUC	Labelled repl.	Unlabelled repl.	Background repl.	Fisher p	O/E lab./unlab.	Robust
Hearing loss								
Exclusion (as primary)	295	0.719	26/44 (59%)	10/51 (20%)	19/158 (12%)	0.242	1.10 / 0.26	36 (10)
Co-medication as stratum	323	0.718	28/43 (65%)	17/57 (30%)	28/158 (18%)	0.060	0.94 / 0.25	45 (17)
Strong ototoxic list only	303	0.714	26/43 (60%)	14/54 (26%)	22/158 (14%)	0.058	1.03 / 0.28	40 (14)
Without reports listing ≥15 drugs	269	0.725	26/41 (63%)	8/48 (17%)	5/159 (3%)	0.002	1.18 / 0.27	34 (8)
Exclusion; Canada from 2004	295	0.719	18/37 (49%)	10/48 (21%)	15/125 (12%)	0.152	1.03 / 0.23	28 (10)
Tinnitus								
Exclusion (as primary)	346	0.776	52/89 (58%)	12/50 (24%)	8/153 (5%)	<0.001	0.70 / 0.21	64 (12)
Co-medication as stratum	434	0.762	59/96 (61%)	20/66 (30%)	15/154 (10%)	<0.001	0.62 / 0.17	79 (20)
Strong ototoxic list only	410	0.769	54/97 (56%)	12/60 (20%)	14/154 (9%)	0.036	0.62 / 0.12	66 (12)
Without reports listing ≥15 drugs	317	0.786	47/86 (55%)	7/45 (16%)	1/153 (1%)	<0.001	0.94 / 0.32	54 (7)
Exclusion; Canada from 2004	346	0.776	41/81 (51%)	14/44 (32%)	8/111 (7%)	<0.001	0.72 / 0.29	55 (14)

Table 3: Previously reported FAERS hearing-loss signals re-tested here (FAERS without Canadian reports). Columns: reported, crude signal, adjusted signal, adjusted and testable in Canada, adjusted and replicated, robust (post hoc).

Study	Authors' label status	Reported	Crude	Adjusted	Testable	Replicated	Robust
Li et al. 2024 (all ages)	labelled	18	18	15	10	10	8
Li et al. 2024 (all ages)	unlabelled	23	16	16	6	2	1
Liu et al. 2025 (children)	labelled	29	16	15	13	11	11
Liu et al. 2025 (children)	unlabelled	28	14	11	9	7	1

duced "replications" for drugs with no known association, and co-medication bias can replicate when crude estimates are compared. After adjusted replication, a review of report provenance and an active-comparator check, 7 of 22 robust unlabelled signals remained (2 to 9 under alternative post-hoc choices and assessors). For tinnitus the same pattern held, but labelled signals also replicated less fully once both databases were adjusted, so the evidence there is weaker.

These results qualify the previous all-drug screens [2–4] rather than contradict them. Their labelled signals largely held; their unlabelled signals mostly did not, and several of the strongest are consistent with how reports were collected rather than with pharmacology. Tafamidis and sacubitril/valsartan are examples: hundreds to thousands of FAERS hearing-loss reports, mostly from US consumers and consistent with manufacturer programmes, and none in Canada. The same mechanism produced spurious Canadian "replications" for biologics. Solicited reports are known to distort disproportionality [20], and their share differs between databases, so report provenance can manufacture both a signal and its replication.

In this study three steps did most of the work: comparing *adjusted* estimates in both databases; profiling each case series for concentration—solicited or consumer reports, a single country or year, very drug-dense reports; and comparing the drug with others used for the same indication. The power-calibrated comparison is simple and avoids the outcome-dependent testability of rules based on observed counts; it remains sensitive to overdispersion, and empirical calibration against negative

Table 4: Robust signals absent from all US labels, with case-level review category: A report cluster; C stimulated or solicited reporting; D indication associated with the event; K established ototoxic class; E none of these (bold). FAERS cases: reports without Canadian origin naming the drug as suspect with the event. Consumer and ≥15 drugs: share of those cases. Hypotheses only, not evidence of causation.

Drug	Event	FAERS cases	FAERS adj. OR (95% CI)	Canada adj. OR (95% CI)	Consumer	≥15 drugs	Cat.
Rofecoxib	HL	243	2.39 (2.02–2.82)	4.42 (2.35–8.31)	7%	14%	C
Tozinameran	HL	57	2.36 (1.76–3.15)	7.04 (5.70–8.69)	72%	26%	C
Elasomeran	HL	20	2.36 (1.44–3.86)	3.30 (2.01–5.44)	75%	35%	C
Etanercept	HL	2095	1.15 (1.10–1.21)	2.28 (1.71–3.05)	49%	3%	D
Insulin Aspart	HL	250	2.27 (2.00–2.58)	2.69 (1.11–6.50)	82%	5%	D
Tofacitinib	HL	1728	3.27 (3.11–3.45)	1.39 (1.05–1.84)	82%	4%	E
Pregabalin	HL	1703	3.25 (3.09–3.43)	2.06 (1.24–3.44)	62%	12%	E
Amlodipine	HL	792	2.39 (2.21–2.57)	2.70 (1.65–4.44)	70%	9%	E
Glycopyrrolate	HL	315	3.08 (2.74–3.45)	4.40 (1.82–10.67)	50%	3%	E
Travoprost	HL	140	3.71 (3.13–4.40)	4.96 (1.58–15.54)	75%	0%	E
Candesartan	Tin	422	11.72 (10.40–13.22)	2.80 (1.68–4.68)	38%	83%	A
Propranolol	Tin	189	1.79 (1.40–2.28)	2.31 (1.03–5.17)	24%	66%	A
Tozinameran	Tin	109	6.53 (5.01–8.50)	10.11 (8.46–12.07)	46%	51%	A
Indapamide	Tin	44	2.05 (1.29–3.26)	3.44 (1.53–7.71)	18%	55%	A
Finasteride	Tin	293	9.73 (8.63–10.95)	3.04 (1.26–7.37)	60%	0%	C
Rofecoxib	Tin	189	2.53 (2.13–3.02)	2.26 (1.12–4.54)	8%	21%	C
Elasomeran	Tin	13	3.62 (1.94–6.74)	3.50 (2.26–5.43)	69%	31%	C
Galcanezumab	Tin	96	1.86 (1.52–2.28)	16.70 (6.73–41.48)	88%	0%	D
Chlorthalidone	Tin	10	1.94 (1.04–3.62)	4.57 (1.70–12.29)	90%	0%	D
Doxycycline	Tin	220	4.54 (3.88–5.31)	2.36 (1.12–4.97)	48%	6%	E
Liothyronine	Tin	40	4.90 (3.56–6.75)	16.73 (5.22–53.59)	78%	32%	E
Paromomycin	Tin	3	27.75 (8.63–89.29)	262.76 (53.09–1300.46)	67%	0%	K

controls [21] or regression-based adjustment for co-reported drugs [22] would be natural extensions. We suggest that screens for hearing outcomes report these provenance indicators for every signal, in line with the case-by-case item of READUS-PV [12, 13].

The remaining candidates are hypotheses. Pregabalin, an $\alpha_2\delta$ ligand, had 1,703 FAERS hearing-loss reports (adjusted OR 3.25 (3.09–3.43)) that replicated in Canada and persisted against other drugs for pain and neuropathy; a case of reversible, dose-related hearing loss has been described [23], but age and diabetes confound [24], and its labels list tinnitus and hyperacusis. For amlodipine, hypertension is associated with hearing loss [25], yet the signal persisted against other antihypertensives, while a large cohort found a *lower* risk of sensorineural hearing loss with calcium-channel blockers [26], which argues against a causal effect. Tofacitinib, travoprost and glycopyrrolate persisted within rheumatoid arthritis, glaucoma and COPD, diseases themselves linked to hearing loss [27–29], so residual confounding by disease severity remains possible. Doxycycline can cause intracranial hypertension, of which pulsatile tinnitus is a symptom [30], although few cases reported it. None of these should change practice without studies that have exposure data.

Limitations. Spontaneous reports lack exposure denominators, and disproportionality measures reporting, not risk or incidence. The label reference set is imperfect: labels are partly informed by spontaneous reports, list rare events without causal assessment, and route-specific labels (for example topical gentamicin) blur ingredient-level status; we used it only to compare methods. Excluding reports with an ototoxic co-medication conditions on a variable that reporters may add after

observing the event, which pushed some label-positive drugs below 1; stratifying instead removed that artefact but left residual co-medication bias, and both gave the same replication contrast. Indications are often missing; unreported co-treatment (for example aminoglycosides in mycobacterial disease) remains; content deduplication was not applied to Canada Vigilance and cannot catch near-duplicate series, which the report-cluster indicator only partly addresses; and combination products contribute to each ingredient (valsartan inherits sacubitril/valsartan reports). Canadian reports were removed from FAERS by occurrence country, or reporter country when missing. The Canada Vigilance documentation does not state that the extract is domestic; restricting it to reports sent directly to Health Canada gave similar results (39/72 (54%) of adjusted signals replicated, background 12/58 (21%)). About 8,100 ingredients were screened without multiplicity correction; the replication and calibration analyses, rather than individual p-values, carry the conclusions, and the calibration's expected counts depend on Poisson assumptions that the background shows to be optimistic. Corrections made after the first results (ingredient synonyms, comparator handling for listed drugs, label sections) changed the primary numbers slightly; they are listed in the protocol file. The solicited-report exclusion, the extended indication filter, adjusted-to-adjusted and calibrated replication, the co-medication variants, the active-comparator check and the case-level review were added after the primary results were seen, partly in response to internal review, and are labelled post hoc. The case-level review was applied by the author and by an AI assessor that used the same profiles and search tools, not by independent clinicians; the two assessments are therefore not fully independent, and the agreement statistic should be read in that light. Within-indication comparisons can be affected by channelling (for example tofacitinib as a second-line drug). FAERS report-source codes that would identify solicited reports directly were not used.

Conclusions. In FAERS, most drug–hearing-loss signals that are not already on a label do not survive adjusted replication in Canada Vigilance and a review of report provenance; disproportionality analysis remains a hypothesis-generating tool. Replication in a second spontaneous-reporting database is informative only when both estimates are adjusted and solicited reports and report clusters are handled explicitly. The few remaining candidates, led by pregabalin, merit study with exposure-based designs.

Statements and declarations

Funding: no funding was received for this study. **Competing interests:** until October 2026 the author was employed as a software engineer at Suno, a company that develops practice-management software for hearing clinics; the company had no role in the design, analysis or reporting of this study, and the author has no financial relationship with manufacturers of the drugs studied. **Ethics approval:** not applicable; the study used public, de-identified data. **Consent to participate and for publication:** not applicable. **Data availability:** FAERS and Canada Vigilance data are publicly available from the FDA and Health Canada; result tables are in the repository below. **Code availability:** code, protocol (`PROTOCOL.md`, with commit history) and

results: https://github.com/olegtropinin/faers-ototoxicity. **Author contributions:** OT conceived and designed the study, directed and checked the analysis, and wrote and approved the manuscript.

References

[1] C. Lanvers-Kaminsky, A. G. am Zehnhoff-Dinnesen, R. Parfitt, and G. Ciarimboli. Drug-induced ototoxicity: Mechanisms, pharmacogenetics, and protective strategies. *Clinical Pharmacology & Therapeutics*, 101(4):491–500, 2017. doi: 10.1002/cpt.603.

[2] Baojian Li, Xiaoling Hu, and Zichen Yue. Drug-induced hearing disorders: a disproportionality analysis of the FAERS database. *Frontiers in Pharmacology*, 15:1480994, 2024. doi: 10.3389/fphar.2024.1480994.

[3] Tingxi Wu, Zhigang Zhao, Peng Wang, Qian Du, Yanfeng Shi, Bin Zhu, Jie Dong, and Dongxuan Li. Drug-induced hearing loss: a real-world pharmacovigilance study using the FDA adverse event reporting system database. *Hearing Research*, 461:109262, 2025. doi: 10.1016/j.heares.2025.109262.

[4] Jinfeng Liu, Junyi Tan, Qinli Xiao, Yingtao Bai, En Chang, Chun Su, Yuxun Wei, Hu Zhong, and Wei Wei. Drug-associated hearing impairment in children: a disproportionality analysis of the FDA adverse event reporting system. *Frontiers in Pharmacology*, 16:1532461, 2025. doi: 10.3389/fphar.2025.1532461.

[5] Nagalakshmi Narasimhaswamy, Shyamala Ganesan, and Basavaraj Poojar. Disproportionality analysis of drug-associated tinnitus safety signals: a 43-year pharmacovigilance study. *Naunyn-Schmiedeberg's Archives of Pharmacology*, 399(12):19173–19186, 2026. doi: 10.1007/s00210-026-05556-7.

[6] Luwei Wang, Yanqin Tang, and Pan Zhuge. Analysis of drug-related tinnitus based on the FDA adverse event reporting system database. *British Journal of Hospital Medicine*, 85(12):1–12, 2024. doi: 10.12968/hmed.2024.0380.

[7] Mizuki Tanaka, Shiori Hasegawa, Satoshi Nakao, Kazuyo Shimada, Ririka Mukai, Kiyoka Matsumoto, and Mitsuhiro Nakamura. Analysis of drug-induced hearing loss by using a spontaneous reporting system database. *PLoS ONE*, 14(10):e0217951, 2019. doi: 10.1371/journal.pone.0217951.

[8] S. Favrelière, P. Delaunay, J. P. Lebreton, F. Rouby, M. Atzenhoffer, C. Lafay-Chebassier, and M. C. Pérault-Pochat. Drug-induced hearing loss: a case/non-case study in the French pharmacovigilance database. *Fundamental & Clinical Pharmacology*, 34(3):397–407, 2020. doi: 10.1111/fcp.12533.

[9] Antoine Pariente, Fleur Gregoire, Annie Fourrier-Reglat, Françoise Haramburu, and Nicholas Moore. Impact of safety alerts on measures of disproportionality in spontaneous reporting databases: the notoriety bias. *Drug Safety*, 30(10):891–898, 2007. doi: 10.2165/00002018-200730100-00007.

[10] Ravi Ramessur and Joel M. Gelfand. Facing our FAERS: Disproportionality analyses, signal detection, and interpretation. *Journal of Investigative Dermatology*, 145(9):2115–2117, 2025. doi: 10.1016/j.jid.2025.06.1579.

[11] R. Harpaz, W. DuMouchel, P. LePendu, A. Bauer-Mehren, P. Ryan, and N. H. Shah. Performance of pharmacovigilance signal-detection algorithms for the FDA adverse event reporting system. *Clinical Pharmacology & Therapeutics*, 93(6):539–546, 2013. doi: 10.1038/clpt.2013.24.

[12] Michele Fusaroli, Francesco Salvo, Bernard Begaud, Thamir M. AlShammari, Andrew Bate, Vera Battini, Andreas Brueckner, Gianmario Candore, Carla Carnovale, Salvatore Crisafulli, Paola Maria Cutroneo, Charles Dolladille, Milou-Daniel Drici, Jean-Luc Faillie, Adam Goldman, Manfred Hauben, Maria Teresa Herdeiro, Olivia Mahaux, Katrin Manlik, François Montastruc, Yoshihiro Noguchi, G. Niklas Norén, Roberta Noseda, Igho J. Onakpoya, Antoine Pariente, Elisabetta Poluzzi, Myriam Salem, Daniele Sartori, Nhung T. H. Trinh, Marco Tuccori, Florence van Hunsel, Eugène van Puijenbroek, Emanuel Raschi, and Charles Khouri. The reporting of a disproportionality analysis for drug safety signal detection using individual case safety reports in pharmacovigilance (READUS-PV): Development and statement. *Drug Safety*, 47(6):575–584, 2024. doi: 10.1007/s40264-024-01421-9.

[13] Michele Fusaroli, Francesco Salvo, Bernard Begaud, Thamir M. AlShammari, Andrew Bate, Vera Battini, Andreas Brueckner, Gianmario Candore, Carla Carnovale, Salvatore Crisafulli, Paola Maria Cutroneo, Charles Dolladille, Milou-Daniel Drici, Jean-Luc Faillie, Adam Goldman, Manfred Hauben, Maria Teresa Herdeiro, Olivia Mahaux, Katrin Manlik, François Montastruc, Yoshihiro Noguchi, G. Niklas Norén, Roberta Noseda, Igho J. Onakpoya, Antoine Pariente, Elisabetta Poluzzi, Myriam Salem, Daniele Sartori, Nhung T. H. Trinh, Marco Tuccori, Florence van Hunsel, Eugène van Puijenbroek, Emanuel Raschi, and Charles Khouri. The REporting of A disproportionality analysis for DrUg safety signal detection using individual case safety reports in PharmacoVigilance (READUS-PV): Explanation and elaboration. *Drug Safety*, 47(6):585–599, 2024. doi: 10.1007/s40264-024-01423-7.

[14] Taha A. Kass-Hout, Zhiheng Xu, Matthew Mohebbi, Hans Nelsen, Adam Baker, Jonathan Levine, Elaine Johanson, and Roselie A. Bright. OpenFDA: an innovative platform providing access to a wealth of FDA's publicly available data. *Journal of the American Medical Informatics Association*, 23(3):596–600, 2016. doi: 10.1093/jamia/ocv153.

[15] Barnet Woolf. On estimating the relation between blood group and disease. *Annals of Human Genetics*, 19(4):251–253, 1955. doi: 10.1111/j.1469-1809.1955.tb01348.x.

[16] Kenneth J. Rothman, Stephan Lanes, and Susan T. Sacks. The reporting odds ratio and its advantages over the proportional reporting ratio. *Pharmacoepidemiology and Drug Safety*, 13 (8):519–523, 2004. doi: 10.1002/pds.1001.

[17] G. Niklas Norén, Andrew Bate, Roland Orre, and I. Ralph Edwards. Extending the methods used to screen the WHO drug safety database towards analysis of complex associations and improved accuracy for rare events. *Statistics in Medicine*, 25(21):3740–3757, 2006. doi: 10.1002/sim.2473.

[18] N. Mantel and W. Haenszel. Statistical aspects of the analysis of data from retrospective studies of disease. *Journal of the National Cancer Institute*, 22(4):719–748, 1959. doi: 10.1093/jnci/22.4.719.

[19] James Robins, Norman Breslow, and Sander Greenland. Estimators of the Mantel-Haenszel variance consistent in both sparse data and large-strata limiting models. *Biometrics*, 42(2): 311–323, 1986. doi: 10.2307/2531052.

[20] Jeremy D. Jokinen, Rosalind J. Walley, Michael W. Colopy, Thomas S. Hilzinger, and Peter Verdru. Pooling different safety data sources: Impact of combining solicited and spontaneous reports on signal detection in pharmacovigilance. *Drug Safety*, 42(10):1191–1198, 2019. doi: 10.1007/s40264-019-00843-0.

[21] Martijn J. Schuemie, Patrick B. Ryan, William DuMouchel, Marc A. Suchard, and David Madigan. Interpreting observational studies: why empirical calibration is needed to correct p-values. *Statistics in Medicine*, 33(2):209–218, 2014. doi: 10.1002/sim.5925.

[22] Ola Caster, G. Niklas Norén, David Madigan, and Andrew Bate. Large-scale regression-based pattern discovery: The example of screening the WHO global drug safety database. *Statistical Analysis and Data Mining*, 3(4):197–208, 2010. doi: 10.1002/sam.10078.

[23] R. Yılmaz, Ş. Türk, R. Reisli, and S. Tuncer Uzun. A probable case of pregabalin-related reversible hearing loss. *Ağrı*, 32(2):103–105, 2020. doi: 10.5505/agri.2018.48753.

[24] C. Horikawa, S. Kodama, S. Tanaka, K. Fujihara, R. Hirasawa, Y. Yachi, H. Shimano, N. Yamada, et al. Diabetes and risk of hearing impairment in adults: a meta-analysis. *The Journal of clinical endocrinology and metabolism*, 98(1):51–8, 2013. doi: 10.1210/jc.2012-2119. PMID 23150692.

[25] X. Jin, X. Xu, J. Wang, X. Liu, X. Deng, and H. Xie. Association between hypertension and hearing loss: a systemic review and meta-analysis. *Frontiers in neurology*, 15:1470997, 2024. doi: 10.3389/fneur.2024.1470997. PMID 39839866.

[26] S. Scott Sutton, Joseph Magagnoli, Tammy H. Cummings, and James W. Hardin. Calcium channel blockers and the risk of sensorineural hearing loss. *Otology & Neurotology*, 43(2): e140–e147, 2022. doi: 10.1097/MAO.0000000000003419.

[27] Nikolaos Chaitidis, Patroklos Theocharis, Charalampos Festas, and Ioanna Aritzi. Association of rheumatoid arthritis with hearing loss: a systematic review and meta-analysis. *Rheumatology International*, 40(11):1771–1779, 2020. doi: 10.1007/s00296-020-04609-1.

[28] Joon Mo Kim, Mi Yeon Lee, Jong Woo Kim, Jungmin Lee, Hyun Ji Kim, Soon Cheol Cha, and Na Rae Kim. Open-angle glaucoma and sensorineural hearing impairment in the Korean population. *Current Eye Research*, 45(8):992–999, 2020. doi: 10.1080/02713683.2020.1712424.

[29] Hyo Geun Choi, Mi Jung Kwon, Ji Hee Kim, Joo-Hee Kim, Yoonjoong Kim, and So Young Kim. Association between COPD and hearing loss. *The Laryngoscope*, 134(11):4716–4722, 2024. doi: 10.1002/lary.31548.

[30] N. S. Ali, B. D. Long, N. F. Manzoor, A. Sismanis, and D. H. Coelho. Doxycycline-induced intracranial hypertension presenting as unilateral pulsatile tinnitus. *Otology & neurotology open*, 3(4):e043, 2023. doi: 10.1097/ONO.0000000000000043. PMID 38516546.

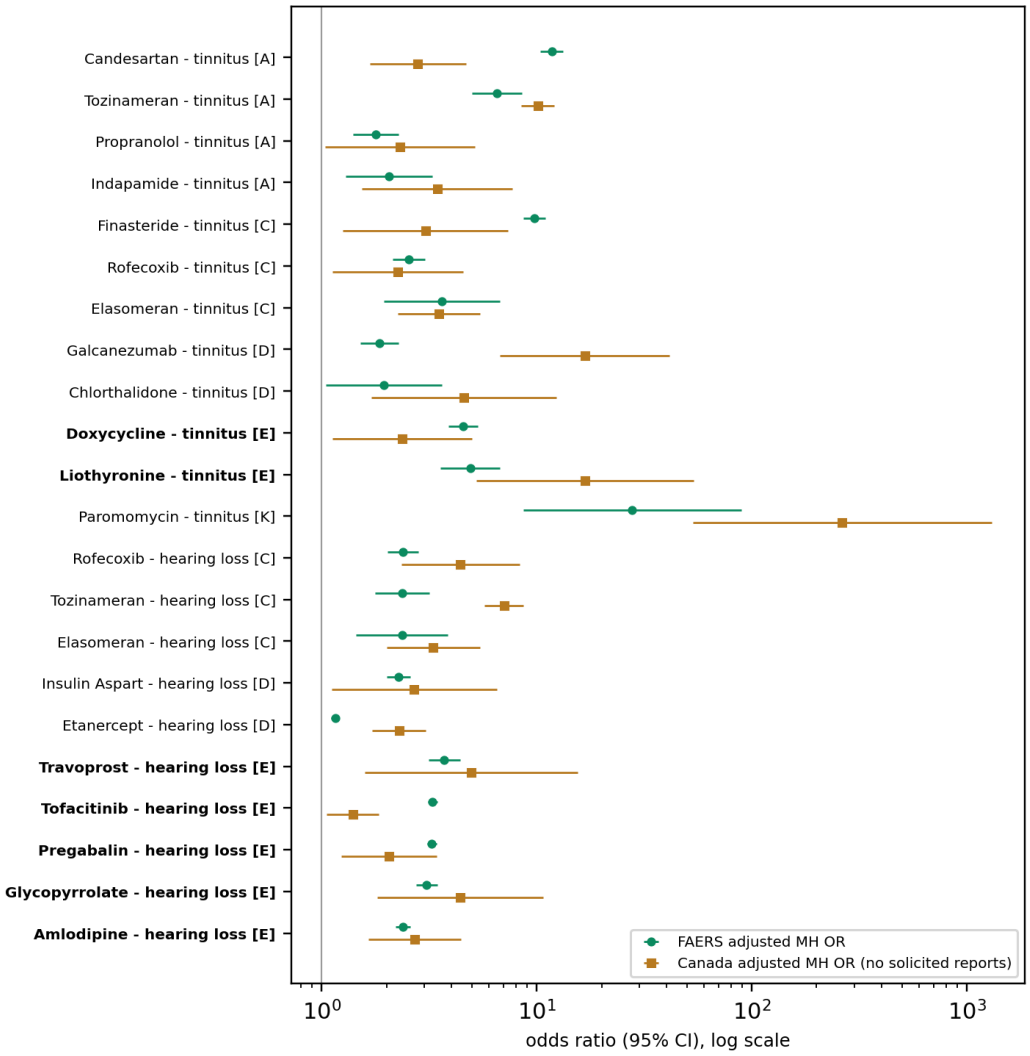


Figure 2: Robust signals absent from all US labels: FAERS and Canada Vigilance adjusted estimates, with case-review category (A report cluster; C stimulated reporting; D indication; K established class; E none, bold).

Supplementary material

Which drug–hearing-loss signals in FAERS survive adjustment and independent replication?

Oleg Tropynin

Table S1. READUS-PV checklist

Item	READUS-PV recommendation (abridged)	Where addressed
1a	Identify the study as a disproportionality analysis; name data type and database(s)	Title
1b	Name the adverse event(s) and/or drug(s)	Title (hearing loss); abstract
2a	Drugs, event, frequency, existing knowledge	Introduction, paragraph 1
2b	Rationale (routine PV, safety profile or pre-specified hypothesis)	Introduction, paragraphs 2–3
2c	Why ICSR databases and disproportionality suit the gap	Introduction, paragraphs 1–3
3	Objectives: event(s), drug(s), reference group, pre-specified hypotheses	Introduction, last paragraph; Methods 2.4
4a	Identify design and data type	Methods 2.1
4b	Outline primary and sensitivity analyses, case-by-case analysis	Methods 2.4–2.8; PROTOCOL.md
5a	Database names, custodians, coverage, thesaurus	Methods 2.1 (FDA, Health Canada; MedDRA PTs)
5b	Extraction dates; pre-processing and exclusions	Methods 2.1–2.2; Results 3.1; Figure S1
6a	Study population and restrictions	Methods 2.3–2.4
6b	Key variables	Methods 2.3
6c	Grouping of drugs/events; ingredients vs trade names; drug role	Methods 2.2–2.3
6d	Additional data sources and how they interact with ICSRs	Methods 2.5 (US labels), 2.8 (PubMed)
7a	Descriptive analyses	Table S4
7b	Disproportionality measures and thresholds	Methods 2.4
7c	Sensitivity analyses and confounding control	Methods 2.4, 2.6, 2.7; Results 3.3–3.4
7d	Variables and criteria for case-by-case analysis	Methods 2.8; Table S3
7e	Statistical methods for other data sources	Methods 2.5 (AUC, bootstrap), 2.6
8a	Number of ICSRs at each stage with reasons	Results 3.1; Figure S1
8b	Demographic and clinical characteristics of cases	Table S4
9	All results with confidence intervals; sensitivity analyses	Tables 1–4, S2, S5, S6; Figures 1–2
10	Case-by-case analysis	Results 3.6; Table 4, S3
11	Key results in context; expected vs emerging signals	Discussion, paragraphs 1–2
12a	External validity	Discussion; Limitations
12b	Relevance to clinical practice	Discussion, paragraph 4
12c	Further study designs	Discussion, paragraph 4; Conclusions
13	Limitations; disproportionality cannot prove causation or measure incidence	Limitations
14a	Funding	Declarations
14b	Conflicts of interest	Declarations
14c	Institutional approval	Declarations (not applicable)
14d	Data, code (software versions) and protocol availability	Methods 2.9; Declarations

Table S2. Established ototoxic drugs: crude and adjusted estimates (hearing loss)

FAERS crude: all reports; FAERS adjusted: Mantel–Haenszel OR in the restricted population (Methods); Canada: crude ROR, all reports.

Ingredient	FAERS cases	FAERS ROR (95% CI)	Adj. cases	FAERS adj. OR (95% CI)	Canada cases	Canada ROR (95% CI)
Gentamicin	315	9.66 (8.63–10.81)	181	11.37 (9.79–13.20)	131	33.60 (27.98–40.34)
Tobramycin	367	5.36 (4.83–5.94)	248	4.82 (4.25–5.47)	31	17.10 (11.89–24.59)
Amikacin	833	16.83 (15.69–18.07)	472	12.70 (11.57–13.95)	35	145.93 (97.41–218.60)
Streptomycin	18	5.76 (3.61–9.20)	12	7.92 (4.47–14.05)	9	6.28 (3.24–12.17)
Cisplatin	1206	4.61 (4.35–4.88)	730	4.27 (3.96–4.59)	138	31.21 (26.13–37.29)
Carboplatin	761	1.96 (1.83–2.11)	251	0.82 (0.72–0.93)	112	19.09 (15.74–23.17)
Oxaliplatin	121	0.44 (0.37–0.53)	106	0.46 (0.38–0.56)	1	0.14 (0.02–1.02)
Vancomycin	365	2.25 (2.03–2.49)	215	2.06 (1.80–2.35)	60	9.25 (7.14–11.98)
Furosemide	412	1.73 (1.57–1.91)	255	1.31 (1.16–1.48)	61	10.18 (7.87–13.16)
Hydroxychloroquine	556	2.10 (1.93–2.28)	327	1.73 (1.55–1.93)	88	2.21 (1.79–2.73)
Quinine	68	5.83 (4.58–7.42)	41	6.10 (4.47–8.31)	4	6.10 (2.27–16.42)
Deferoxamine	51	8.97 (6.78–11.86)	50	11.78 (8.87–15.65)	34	36.29 (25.40–51.86)
Sildenafil	894	3.99 (3.73–4.26)	725	3.93 (3.65–4.24)	32	4.19 (2.95–5.94)
Tadalafil	720	4.52 (4.20–4.87)	594	4.22 (3.89–4.58)	35	4.19 (3.00–5.85)
Vardenafil	6	1.31 (0.59–2.93)	4	0.93 (0.35–2.49)	4	17.01 (6.21–46.61)
Teprotumumab	1133	59.90 (56.11–63.95)	859	59.76 (55.40–64.47)	1	–

Figure S1. Report flow

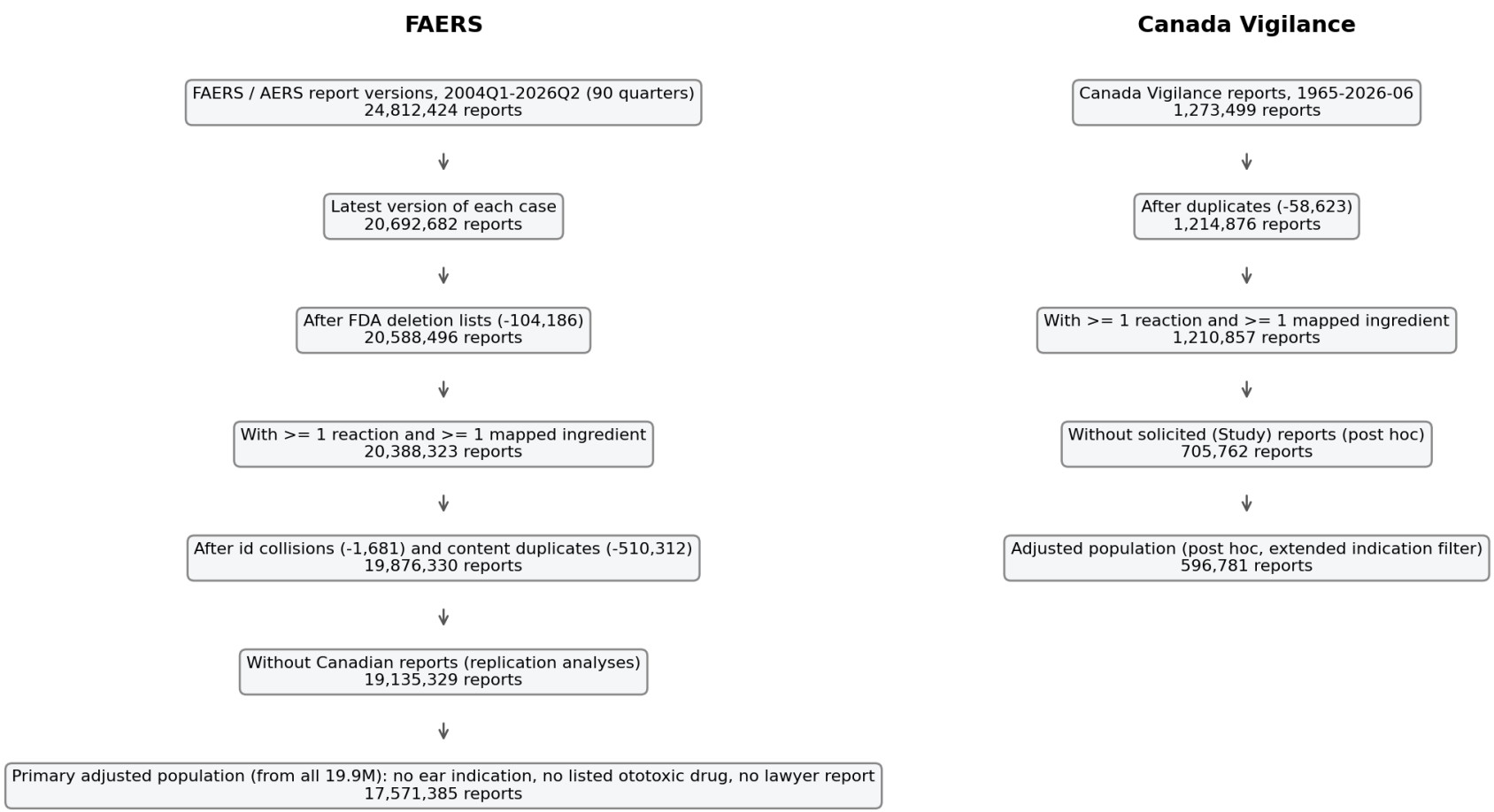


Table S3. Case-level review of robust signals absent from all US labels (post hoc)

Rubric, first applicable category: K established ototoxic class or association documented by a prospective/cohort series; A report cluster ($\geq$50% of cases in reports listing $\geq$15 drugs); B $\geq$50% of cases co-name an established ototoxic drug; C stimulated or solicited reporting; D main indication associated with the event; E none. Cat. 1: author; Cat. 2: second, AI-based assessor, blind to Cat. 1, before the active-comparator rule (agreement 19/22, $\kappa = 0.82$). Category D also requires the active-comparator (within-indication) ROR lower bound ≤ 1. Profiles: FAERS cases without Canadian reports.

Drug	Event	Cat. 1	Cat. 2	Cases	Top country	Top year	Consumer	Median drugs	≥15 drugs	Ototoxic co-drug	Within-indication ROR	Note
Rofecoxib	HL	C	C	243	US 64%	2006 42%	7%	6	14%	33%		litigation-era reporting (2004-2
Tozinameran	HL	C	K	57	US 40%	2022 28%	72%	6	26%	5%		mass vaccination; cohort evic conflicting (Israel SIR 1.35, P 35201275; Denmark null, P 37288514)
Elasomeran	HL	C	C	20	US 85%	2023 35%	75%	8	35%	20%		mass vaccination; media atte to hearing effects
Etanercept	HL	D	D	2095	US 80%	2016 20%	49%	1	3%	4%	0.93 (0.87-1.00)	rheumatoid arthritis assoc with SNHL (PMID 32488 used for autoimmune inner disease
Insulin Aspart	HL	D	D	250	US 85%	2025 17%	82%	2	5%	2%	1.17 (1.00-1.37)	diabetes associated with he impairment (meta-analysis, P 23150692)
Tofacitinib	HL	E	D	1728	US 94%	2022 19%	82%	1	4%	7%	2.24 (2.06-2.45)	persists against other RA d 82 % consumer reports; rhe toid arthritis associated with S (PMID 32488430)
Pregabalin	HL	E	E	1703	US 86%	2021 23%	62%	2	12%	11%	3.27 (3.04-3.52)	alpha-2-delta ligand; one report of reversible loss (P 32297964); diabetes/neurop and age (median 71) confo labels list tinnitus and hyperac not hearing loss
Amlodipine	HL	E	D	792	US 29%	2021 15%	70%	4	9%	14%	1.89 (1.69-2.10)	labels list tinnitus; VA cohort f lower SNHL risk with calc channel blockers (PMID 34855 hypertension associated with ing loss (meta-analysis, P 39839866)
Glycopyrrolate	HL	E	D	315	US 75%	2025 27%	50%	3	3%	4%	1.65 (1.27-2.14)	persists against other COPD d COPD associated with hearing (PMID 38828668)
Travoprost	HL	E	D	140	US 61%	2019 23%	75%	1	0%	3%	2.43 (1.80-3.29)	persists against other glauc drugs; median age 80; coma associated with SNHL (P 31935130)
Candesartan	Tin	A	A	422	GB 80%	2023 54%	38%	42	83%	34%		83% of cases in reports listing 15 drugs
Propranolol	Tin	A	A	189	PT 50%	2024 21%	24%	71	66%	66%		66% of cases in reports listing 15 drugs
Tozinameran	Tin	A	A	109	GB 52%	2023 38%	46%	28	51%	48%		51% of cases in reports listing 15 drugs; mass vaccination; hort evidence conflicting (Israe 1.35, PMID 35201275; Den null, PMID 37288514)
Indapamide	Tin	A	A	44	PL 50%	2018 23%	18%	15.5	55%	59%		55% of cases in reports listing 15 drugs
Finasteride	Tin	C	C	293	GB 38%	2024 17%	60%	1	0%	1%		post-finasteride syndrome advo reporting

Rofecoxib	Tin	C	C	189	US 58%	2006 47%	8%	6	21%	26%		litigation-era reporting (2004-2
Elasomeran	Tin	C	C	13	US 54%	2023 38%	69%	8	31%	23%		mass vaccination; media atte to hearing effects
Galcanezumab	Tin	D	E	96	US 94%	2020 28%	88%	1	0%	1%	0.82 (0.63-1.06)	migraine associated with t tus (systematic review, P 36459425; a meta-analysis f low evidence, PMID 36380120
Chlorthalidone	Tin	D	E	10	US 70%	2025 20%	90%	4	0%	0%	1.04 (0.43-2.51)	hypertension associated tinnitus (meta-analysis, P 26881064)
Doxycycline	Tin	E	E	220	US 43%	2024 13%	48%	3	6%	23%	3.63 (2.98-4.42)	pulsatile tinnitus can accom labelled drug-induced intracr hypertension (PMID 38516 but only 15/220 cases co-r intracranial hypertension or p loedema
Liothyronine	Tin	E	D	40	US 57%	2023 35%	78%	5.5	32%	2%	2.10 (1.31-3.36)	persists against other hyp roidism drugs; 32 % drug- reports (UK cluster overlap); pothyroidism associated with t tus (PMID 35233053)
Paromomycin	Tin	K	K	3	US 67%	2017 33%	67%	1	0%	0%		on the a-priori ototoxic list (ar glycoside); oral aminoglyc poorly absorbed; 3 cases

Table S4. Characteristics of FAERS hearing-loss and tinnitus reports and all other reports (READUS-PV item 8b)

	Hearing loss	Tinnitus	All other reports
Reports, n	83,883	42,900	19,756,269
Age, median (years)	67	55	58
Age			
<18 (%)	2.4	1.2	3.5
18-44 (%)	6.9	16.3	12.5
45-64 (%)	15.6	30.0	20.2
≥65 (%)	30.6	17.4	20.7
unknown (%)	44.5	35.1	43.1
Sex			
female (%)	52.9	58.3	52.7
male (%)	39.3	32.2	34.3
unknown (%)	7.8	9.5	13.0
Reporter			
consumer (%)	60.0	53.7	44.2
physician (%)	16.2	16.9	21.6

pharmacist (%)	3.2	4.7	6.1
other health professional (%)	12.6	14.2	18.6
lawyer (%)	0.8	1.8	2.5
unknown (%)	7.2	8.7	6.9
Country			
United States (%)	69.9	64.0	71.3
other (%)	28.6	32.8	27.0
unknown (%)	1.5	3.2	1.7
FDA receipt			
2004-2012 (%)	13.6	21.2	15.6
2013-2018 (%)	30.7	34.0	31.8
2019-2026 (%)	55.7	44.8	52.6

Table S5. All robust signals (post hoc: adjusted FAERS signal replicated adjusted-to-adjusted in Canada Vigilance without solicited reports)

FAERS: adjusted MH OR (extended indication filter, Canadian reports excluded); Canada: adjusted MH OR; Label: event on any US label of a product containing the drug.

Ingredient	Event	FAERS adj. cases	FAERS adj. OR (95% CI)	Canada adj. cases	Canada adj. OR (95% CI)	Label
Amikacin	HL	454	12.71 (11.55–13.98)	20	286.54 (160.73–510.83)	yes
Amlodipine	HL	675	2.39 (2.21–2.57)	16	2.70 (1.65–4.44)	no
Amphotericin B	HL	43	1.53 (1.14–2.07)	3	6.30 (2.01–19.75)	yes
Azithromycin	HL	404	5.15 (4.66–5.68)	30	11.05 (7.65–15.96)	yes
Bupropion	HL	219	1.15 (1.01–1.32)	21	2.46 (1.59–3.80)	yes
Ciprofloxacin	HL	400	2.94 (2.66–3.24)	35	4.47 (3.19–6.27)	yes
Cisplatin	HL	688	4.30 (3.98–4.64)	47	20.99 (15.49–28.45)	yes
Clarithromycin	HL	196	3.40 (2.95–3.91)	21	5.45 (3.52–8.43)	yes
Deferasirox	HL	290	4.02 (3.58–4.52)	6	4.29 (1.91–9.66)	yes
Deferoxamine	HL	35	9.60 (6.85–13.45)	24	81.44 (51.89–127.82)	yes
Elasomeran	HL	16	2.36 (1.44–3.86)	16	3.30 (2.01–5.44)	no
Erythromycin	HL	57	2.29 (1.76–2.97)	20	7.74 (4.90–12.22)	yes
Etanercept	HL	2003	1.15 (1.10–1.21)	49	2.28 (1.71–3.05)	no
Ethacrynic Acid	HL	4	13.94 (5.07–38.28)	5	109.98 (41.12–294.17)	yes
Furosemide	HL	209	1.19 (1.04–1.36)	14	4.42 (2.59–7.53)	yes
Gentamicin	HL	168	11.96 (10.24–13.97)	90	65.05 (50.65–83.55)	yes
Glycopyrrolate	HL	303	3.08 (2.74–3.45)	5	4.40 (1.82–10.67)	no
Hydroxychloroquine	HL	133	1.80 (1.51–2.13)	63	2.03 (1.57–2.62)	yes
Insulin Aspart	HL	243	2.27 (2.00–2.58)	5	2.69 (1.11–6.50)	no
Moxifloxacin	HL	132	2.12 (1.78–2.51)	23	6.87 (4.52–10.43)	yes
Neomycin	HL	52	4.97 (3.77–6.54)	6	8.70 (3.85–19.66)	yes
Peginterferon Alfa-2B	HL	64	1.54 (1.20–1.96)	7	4.57 (2.15–9.70)	yes
Piroxicam	HL	15	2.20 (1.32–3.65)	3	5.65 (1.80–17.75)	yes
Pregabalin	HL	1497	3.25 (3.09–3.43)	15	2.06 (1.24–3.44)	no
Ramipril	HL	82	1.27 (1.02–1.58)	10	2.34 (1.25–4.37)	yes
Ribavirin	HL	323	1.25 (1.12–1.40)	11	1.98 (1.09–3.61)	yes

Rofecoxib	HL	143	2.39 (2.02–2.82)	10	4.42 (2.35–8.31)	no
Sildenafil	HL	705	4.04 (3.74–4.35)	23	5.44 (3.57–8.28)	yes
Streptomycin	HL	12	9.18 (5.17–16.29)	6	8.80 (3.89–19.93)	yes
Tadalafil	HL	587	4.42 (4.07–4.79)	17	4.43 (2.71–7.22)	yes
Terbinafine	HL	99	3.29 (2.70–4.01)	5	3.09 (1.28–7.48)	yes
Tobramycin	HL	232	4.82 (4.23–5.49)	17	23.12 (14.12–37.86)	yes
Tofacitinib	HL	1608	3.27 (3.11–3.45)	53	1.39 (1.05–1.84)	no
Tozinameran	HL	46	2.36 (1.76–3.15)	109	7.04 (5.70–8.69)	no
Travoprost	HL	135	3.71 (3.13–4.40)	3	4.96 (1.58–15.54)	no
Vancomycin	HL	209	2.21 (1.93–2.54)	42	12.16 (8.87–16.67)	yes
Acetazolamide	Tin	12	2.96 (1.68–5.23)	3	5.11 (1.63–16.07)	yes
Amikacin	Tin	233	17.86 (15.66–20.37)	3	21.25 (6.56–68.80)	yes
Amitriptyline	Tin	72	1.68 (1.33–2.12)	11	1.96 (1.08–3.56)	yes
Amlodipine	Tin	445	3.06 (2.79–3.36)	16	2.50 (1.52–4.10)	yes
Aspirin	Tin	615	3.01 (2.78–3.26)	118	7.51 (6.19–9.10)	yes
Atenolol	Tin	83	2.91 (2.35–3.62)	8	3.91 (1.94–7.88)	yes
Azithromycin	Tin	257	5.76 (5.09–6.52)	22	6.00 (3.92–9.19)	yes
Bisoprolol	Tin	321	13.85 (12.39–15.49)	4	3.29 (1.23–8.83)	yes
Bupropion	Tin	1058	7.90 (7.42–8.41)	59	4.51 (3.47–5.87)	yes
Buspirone Hydrochloride	Tin	44	3.84 (2.85–5.16)	4	8.70 (3.20–23.61)	yes
Candesartan	Tin	277	11.72 (10.40–13.22)	15	2.80 (1.68–4.68)	no
Celecoxib	Tin	154	1.67 (1.42–1.95)	20	1.70 (1.09–2.65)	yes
Chlorthalidone	Tin	10	1.94 (1.04–3.62)	4	4.57 (1.70–12.29)	no
Choline	Tin	4	23.45 (8.56–64.21)	4	3.98 (1.48–10.70)	yes
Ciprofloxacin	Tin	969	12.72 (11.92–13.57)	52	4.98 (3.77–6.58)	yes
Cisplatin	Tin	233	2.33 (2.05–2.65)	17	5.81 (3.58–9.43)	yes
Citalopram	Tin	179	2.37 (2.05–2.75)	12	1.77 (1.00–3.13)	yes
Clonazepam	Tin	254	2.99 (2.64–3.38)	16	3.58 (2.18–5.88)	yes
Cyclobenzaprine	Tin	33	1.57 (1.12–2.22)	4	2.96 (1.10–7.95)	yes
Desvenlafaxine	Tin	226	4.74 (4.15–5.40)	9	4.16 (2.14–8.06)	yes
Dexchlorpheniramine	Tin	8	4.34 (2.16–8.71)	5	2.80 (1.16–6.78)	yes
Diltiazem	Tin	48	1.70 (1.28–2.26)	11	4.05 (2.23–7.38)	yes
Doxycycline	Tin	159	4.54 (3.88–5.31)	7	2.36 (1.12–4.97)	no
Duloxetine	Tin	1718	12.40 (11.80–13.04)	32	4.16 (2.92–5.92)	yes
Elasomeran	Tin	10	3.62 (1.94–6.74)	21	3.50 (2.26–5.43)	no
Felodipine	Tin	220	73.24 (63.47–84.52)	3	5.34 (1.70–16.75)	yes
Finasteride	Tin	286	9.73 (8.63–10.95)	5	3.04 (1.26–7.37)	no
Gabapentin	Tin	341	2.02 (1.82–2.25)	25	4.21 (2.83–6.28)	yes
Galcanezumab	Tin	94	1.86 (1.52–2.28)	5	16.70 (6.73–41.48)	no
Gentamicin	Tin	45	5.75 (4.28–7.72)	23	8.48 (5.55–12.97)	yes
Ibuprofen	Tin	289	1.51 (1.34–1.69)	36	2.89 (2.07–4.04)	yes
Indapamide	Tin	18	2.05 (1.29–3.26)	6	3.44 (1.53–7.71)	no
Indomethacin	Tin	15	2.54 (1.53–4.22)	9	2.68 (1.39–5.20)	yes
Irbesartan	Tin	73	3.15 (2.50–3.97)	6	2.26 (1.01–5.07)	yes
Isotretinoin	Tin	212	2.69 (2.35–3.08)	10	3.17 (1.69–5.94)	yes
Lidocaine	Tin	90	1.67 (1.35–2.05)	12	2.85 (1.61–5.04)	yes
Liothyronine	Tin	38	4.90 (3.56–6.75)	3	16.73 (5.22–53.59)	no
Mefloquine	Tin	85	27.80 (22.24–34.74)	8	14.52 (7.08–29.79)	yes
Minocycline	Tin	49	4.01 (3.02–5.31)	7	3.49 (1.65–7.36)	yes
Moxifloxacin	Tin	151	3.51 (2.99–4.13)	21	4.65 (3.01–7.18)	yes
Naproxen	Tin	445	3.05 (2.77–3.35)	23	2.42 (1.60–3.66)	yes
Nifedipine	Tin	24	1.51 (1.01–2.26)	7	2.95 (1.40–6.22)	yes
Norfloxacin	Tin	7	4.63 (2.19–9.75)	4	6.03 (2.23–16.29)	yes

Nortriptyline	Tin	29	2.80 (1.94–4.04)	6	3.96 (1.76–8.87)	yes
Paromomycin	Tin	3	27.75 (8.63–89.29)	3	262.76 (53.09–1300.46)	no
Paroxetine	Tin	478	5.19 (4.74–5.69)	38	5.12 (3.70–7.11)	yes
Penicillin G Procaine	Tin	5	7.93 (3.27–19.22)	5	10.12 (4.12–24.86)	yes
Perindopril Erbumine	Tin	6	2.33 (1.05–5.20)	12	3.59 (2.02–6.37)	yes
Propranolol	Tin	65	1.79 (1.40–2.28)	6	2.31 (1.03–5.17)	no
Quinine	Tin	35	7.11 (5.09–9.94)	10	23.00 (11.99–44.15)	yes
Rofecoxib	Tin	131	2.53 (2.13–3.02)	8	2.26 (1.12–4.54)	no
Sildenafil	Tin	214	2.28 (1.99–2.61)	12	2.59 (1.46–4.59)	yes
Streptomycin	Tin	3	3.37 (1.08–10.51)	14	10.88 (6.31–18.74)	yes
Telmisartan	Tin	58	3.22 (2.49–4.17)	8	2.02 (1.00–4.07)	yes
Terbinafine	Tin	58	2.95 (2.28–3.82)	7	2.87 (1.36–6.07)	yes
Tobramycin	Tin	144	6.04 (5.12–7.12)	7	6.50 (3.06–13.80)	yes
Topiramate	Tin	193	2.68 (2.32–3.09)	9	1.99 (1.03–3.84)	yes
Tozinameran	Tin	56	6.53 (5.01–8.50)	171	10.11 (8.46–12.07)	no
Tramadol	Tin	183	1.46 (1.26–1.69)	10	1.99 (1.07–3.72)	yes
Trazodone Hydrochloride	Tin	59	1.33 (1.03–1.72)	12	4.63 (2.61–8.20)	yes
Valsartan	Tin	347	1.43 (1.29–1.59)	14	2.23 (1.31–3.80)	yes
Vancomycin	Tin	90	1.68 (1.37–2.07)	9	2.03 (1.05–3.92)	yes
Venlafaxine Hydrochloride	Tin	603	4.96 (4.57–5.38)	6	3.15 (1.40–7.05)	yes
Vortioxetine	Tin	84	2.74 (2.21–3.40)	4	3.06 (1.14–8.20)	yes

Table S6. Signals of previous screens available only from abstracts

Study	Scope	Reported	Crude	Adjusted	Replicated	Robust	Robust drugs
narasimhaswamy2026	tinnitus	4	0	0	0	0	
narasimhaswamy2026	tinnitus	3	2	1	1	1	IBUPROFEN
wang2024	tinnitus	3	2	2	2	2	CIPROFLOXACIN;DULOXETINE
wu2025	hearing_loss	5	3	2	0	0	